# Few-Trial Damage Recovery with Quality-Diversity Repertoires

Repertoires built with and without reinforcement learning, adapted with Intelligent Trial & Error

Anaïs Azouaoui

https://github.com/ananalab/qd-damage-recovery

**Abstract**

A legged robot that loses the use of a leg should find a new gait with as few physical trials as possible. Intelligent Trial & Error (ITE) does so by searching, with Bayesian optimisation, a repertoire of diverse gaits learned beforehand in simulation. Recent Quality-Diversity (QD) algorithms build such repertoires with the help of reinforcement-learning (RL) critics and reach much higher performance, but it is not known whether they keep the diversity that makes a repertoire useful after damage. On the Brax Ant, we build repertoires at equal budget with MAP-Elites, PGA-ME and DCRL-ME (3 seeds each), measure their resilience to 12 damage scenarios, and adapt them with ITE, simple baselines and RL (TD3), counting episodes on the damaged robot. Repertoires built with RL are as resilient as those of MAP-Elites in relative terms and keep much better gaits in absolute terms. ITE needs a median of 5 trials and recovers 33% of the progress of the best intact gait (against 18% without adaptation), but it is barely better than trying gaits in decreasing order of intact performance, even with a descriptor that models the damage three times better or with a gap between simulator and robot. Re-training with TD3 needs hundreds of episodes once its critic is warmed up. A TD3 policy trained in simulation on random damages, however, matches ITE on the best repertoires without any trial, as long as the damage belongs to its training distribution.

## Contents

**1 Introduction** **2**

**2 Background and related work** **3**
- 2.1 Quality-Diversity and MAP-Elites . . . 3
- 2.2 QD with reinforcement learning . . . 3
- 2.3 Damage recovery with repertoires . . . 3
- 2.4 Adaptation without a repertoire . . . 4

**3 Experimental setup** **4**
- 3.1 Robot, task and behaviour descriptor . . . 4
- 3.2 Algorithms and budget . . . 4
- 3.3 Damage scenarios . . . 4
- 3.4 Performance scale . . . 5

**4 Methods** **5**
- 4.1 Oracle and resilience . . . 5
- 4.2 Intelligent Trial & Error . . . 5
- 4.3 Baselines . . . 6
- 4.4 Statistics . . . 6
- 4.5 RL re-training . . . 6

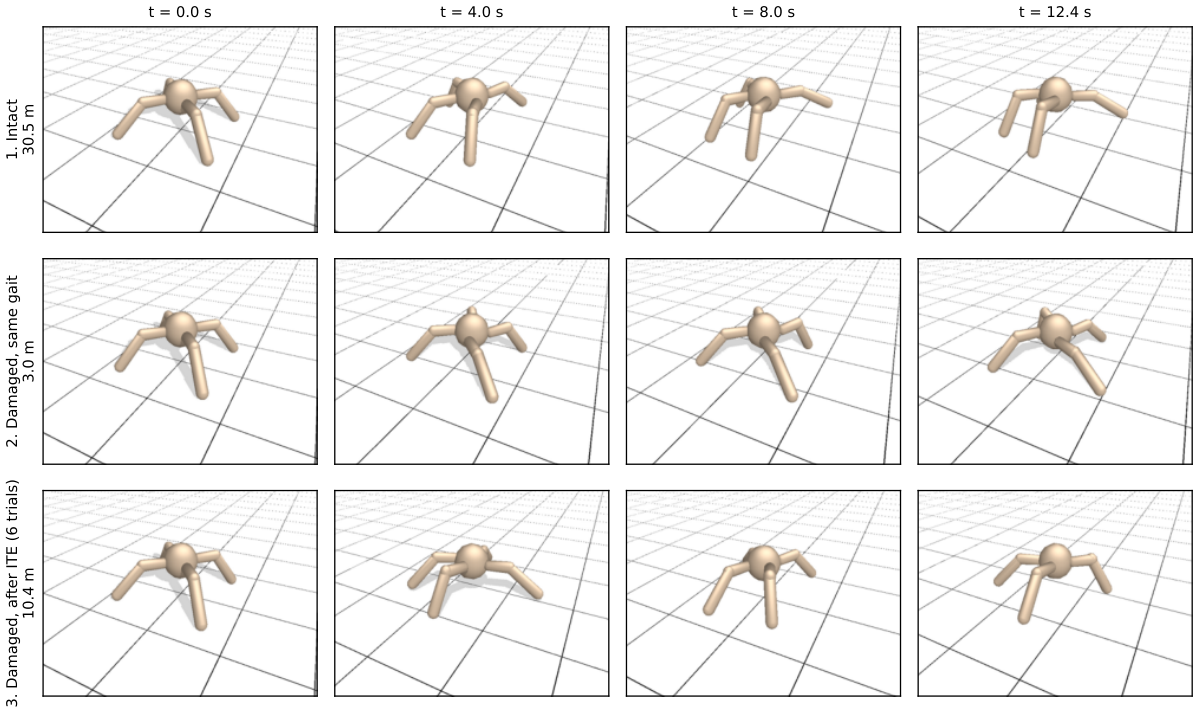



**Figure 1:** The simulated quadruped with its best gait (top), the same gait after the front-right leg is paralysed (middle), and the gait found by ITE in a few trials (bottom). Distances travelled in 12.5 s on the left.

**5 Results** **7**
5.1 Repertoires on the intact robot . . . . . 7
5.2 Resilience to damage (Q1) . . . . . 7
5.3 Adaptation in a few trials (Q2) . . . . . 8
5.4 Comparison with RL re-training (Q3) . . . . . 11
5.5 A damage-robust policy . . . . . 11
5.6 Bonus: searching the continuous descriptor space . . . . . 12
5.7 Robustness of the conclusions . . . . . 12

**6 Discussion** **14**

**7 Limitations** **15**

**8 Conclusion and future work** **15**

**A Reproducibility** **16**

# 1 Introduction

Animals keep walking after an injury because they already know many ways to move: they limp, shift their weight, use another leg more. Cully et al. [5] gave robots the same ability. *Before* any damage, a large repertoire of diverse gaits is learned in simulation with MAP-Elites [12]; *after* damage, the robot runs Intelligent Trial & Error (ITE): it tests a promising gait, observes how well it works, and updates a Gaussian-process model of the performance of *all similar* gaits, so that each trial is informative about many of them. A damaged hexapod recovered in less than two minutes.

Since then, QD algorithms have been combined with deep RL. PGA-ME [14] improves half of the offspring with the gradient of a TD3 critic [9]; DCRL-ME [6] conditions critic and actor on the targeted behaviour descriptor, so that RL can improve a gait *without changing its style*. These methods produce much better gaits. Whether they also produce repertoires that are useful after damage is less clear: RL pushes towards high performance, possibly at the cost of the diversity that adaptation needs. Moreover, adaptation has mostly been evaluated with an *oracle* that tries every skill on the damaged robot [2], which says nothing about the number of trials a real robot would need.

**Summary of results** (i) At equal budget, DCRL-ME covers the most cells (83.8%) and PGA-ME finds the fastest gaits (best fitness 1922). The *relative* resilience to damage does not differ significantly between algorithms, but the best gait left after damage is much better in repertoires built with RL. (ii) ITE stops after a median of 5 trials and recovers 33% of the intact progress, against 18% for the best intact gait; however, a model-free strategy that tries gaits in decreasing order of intact performance (top-$k$) does almost as well: ITE is slightly ahead after three trials, mostly on DCRL-ME repertoires, and no longer after five; a better descriptor or a reality gap do not change this. (iii) RL re-training needs hundreds of episodes on the damaged robot, and fine-tuning the intact policy only works once its critic is warmed up or pre-trained; a damage-robust TD3 policy, trained in simulation, matches ITE without any trial on damages it was trained for. (iv) Searching the continuous descriptor space with the DCRL-ME conditioned actor, instead of the repertoire cells, is clearly worse.

**Questions**

**Q1.** At equal budget, are repertoires built with RL (PGA-ME, DCRL-ME) more or less *resilient* to damage than those of MAP-Elites?

**Q2.** How many *trials* does ITE need on each kind of repertoire, compared with simpler strategies?

**Q3.** How does this compare with re-training a policy by RL on the damaged robot, counted in episodes?

# 2 Background and related work

## 2.1 Quality-Diversity and MAP-Elites

A QD algorithm searches for a *collection* of solutions that are both diverse and high-performing. MAP-Elites [12] discretises a user-defined *behaviour descriptor* space into cells and keeps, in each cell, the best solution found (its *elite*). At each iteration, elites are selected at random, mutated, evaluated, and inserted in their cell if they beat its current occupant.

## 2.2 QD with reinforcement learning

Mutations are blind. PGA-ME [14] trains a TD3 critic on all transitions collected by the population and uses its gradient to improve half of the offspring. DCG-ME and DCRL-ME [6] condition the critic on a target descriptor, so that the gradient improves a policy while keeping its behaviour, and also train a descriptor-conditioned actor whose outputs are added to the repertoire. Chalumeau et al. [2] compared QD methods with skill-discovery RL (DIAYN, DADS, SMERL) on few-shot adaptation tasks, including a leg dysfunction on Ant: after perturbation, every skill is evaluated and the best one is kept. QD methods were more robust to large perturbations, and PGA-ME adapted better than MAP-Elites.

### 2.3 Damage recovery with repertoires

ITE [5] uses the M-BOA algorithm described in Section 4.2. Follow-ups removed the need for resets [4], selected among several priors [10], or used hierarchical repertoires [1]. We keep the original algorithm and study the effect of *how the repertoire was built*. Noisy evaluations, which make elites look better than they are, are the topic of uncertain QD [8].

### 2.4 Adaptation without a repertoire

Other approaches prepare a single policy for a family of conditions. Domain randomisation trains it on randomised dynamics so that it works zero-shot [15, 16]; strategy optimisation searches a small latent space of a conditioned policy on the new robot [17]; meta-learning trains a policy, or a model, to adapt from a few gradient steps or a few seconds of data [7, 13]; RMA infers the current dynamics online from the recent history [11]. We compare ITE with the simplest member of this family, a TD3 policy trained on random damages (Section 5.5), and leave meta-learned baselines for future work.

## 3 Experimental setup

### 3.1 Robot, task and behaviour descriptor

We use the Brax Ant (`ant_uni` in QDax [3]): a torso and four legs, each with a hip and an ankle actuator (8 actions). The reward is the forward velocity plus a survival bonus, minus control and contact costs, over episodes of 250 steps (12.5 s). As in the DCRL-ME notebook, rewards are shifted by a constant and clipped at zero for all algorithms. The descriptor is the fraction of time each foot touches the ground, $\mathbf{d} \in [0, 1]^4$; the repertoire is a centroidal Voronoi tessellation of 1024 cells.

**Leg mapping** The mapping from actions to legs was checked in three independent ways (geometry of the bodies at reset, relative motion of each foot when a single action is applied, and video): actions $(0, 1)$, $(2, 3)$, $(4, 5)$, $(6, 7)$ drive the hip and ankle of the front-left (FL), hind-left (HL), hind-right (HR) and front-right (FR) legs.

### 3.2 Algorithms and budget

The three algorithms share the environment, the policy architecture (MLP, two hidden layers of 128 units, tanh output), the repertoire, and the number of evaluations per iteration (256); hyper-parameters specific to each algorithm are those of the official QDax notebooks (Table 1). Each run lasts 2000 iterations, i.e. 512,256 episodes; 3 seeds per algorithm. A test checks that the three configurations only differ by their emitter.

**What "equal budget" means** The budget is equal in *episodes*, which is what a robot would pay, not in computation: per iteration, PGA-ME also makes 300 critic and 100 policy-gradient steps and DCRL-ME 3000 and 150 (Table 1), so DCRL-ME makes about eight times more gradient steps than PGA-ME (82 against 52 minutes per run on a T4, against 8 for MAP-Elites). The hyper-parameters of each algorithm come from the official notebooks, which use other tasks (Walker2d, Ant-omni) and were not tuned for `ant_uni`; the ranking between PGA-ME and DCRL-ME may depend on this choice.

### 3.3 Damage scenarios

A damage multiplies the actions of some joints by a factor $\lambda$ ($\lambda = 0$: paralysed, $\lambda = 0.5$: weakened). Since the torque of an Ant actuator is proportional to its action, this is equivalent to

**Table 1:** Main hyper-parameters (full configuration in `configs/`).

| | MAP-Elites | PGA-ME | DCRL-ME |
|---|---|---|---|
| Offspring per iteration | 256 mutations | 128 mutations + 128 PG | 128 + 64 PG + 64 actor |
| Iso/line variation $\sigma$ | 0.005 / 0.05 | 0.005 / 0.05 | 0.005 / 0.05 |
| Critic | – | TD3, $256 \times 256$ | descriptor-cond., $256 \times 256$ |
| Critic steps / iteration | – | 300 | 3000 |
| PG steps / iteration | – | 100 | 150 |

scaling its strength, as in Chalumeau et al. [2]. We use 12 scenarios: each leg paralysed, each leg weakened, two legs paralysed (FL+HL, same side; FL+FR, both front legs), and one ankle paralysed (FL, HR).

### 3.4 Performance scale

Episodes are noisy (random initial state), and a robot that does not move still collects the survival bonus: its return $F_0$ was measured at 4.23 per step (1058.7 per episode). We report

$$v = \frac{F - F_0}{S}, \qquad S = 1.2 \text{ per step (300 per episode)},$$

so that $v \approx 0$ for a standing robot and $v \approx 1$ for $\approx$15 m in 12.5 s, or percentages of the progress $F - F_0$ of the best intact gait. With this scale, the variance of a single episode's performance is $\approx 10^{-3}$, the value used by Cully et al. [5].

## 4 Methods

### 4.1 Oracle and resilience

For each repertoire and damage, every policy is evaluated on 4 episodes. The best mean is the *oracle* performance, an upper bound for any method that picks a gait from the repertoire (it needs 4 episodes for each of the 731–862 policies of a repertoire, for every damage). The *resilience* of a repertoire to a damage is the fraction of the intact oracle progress still reachable: $(F^\star_{\text{damaged}} - F_0)/(F^\star_{\text{intact}} - F_0)$.

**Selection bias of the oracle** The oracle is the maximum of noisy means over several hundred cells, so it is biased upwards, and possibly more for repertoires with more cells. As a check, the cell selected by the oracle is re-evaluated on 16 fresh episodes, which removes the selection bias (Section 5.7).

### 4.2 Intelligent Trial & Error

Let $P(\mathbf{d}_i)$ be the prior performance of cell $i$ on the intact robot. ITE models the residual between observed and prior performance with a Gaussian process with a Matérn-5/2 kernel on descriptors:

$$\mu_t(\mathbf{d}) = P(\mathbf{d}) + \mathbf{k}(\mathbf{d})^\top (K + \sigma_n^2 I)^{-1}(\mathbf{y} - P(\mathbf{D}_t)),$$
$$\sigma_t^2(\mathbf{d}) = \sigma_f^2 - \mathbf{k}(\mathbf{d})^\top (K + \sigma_n^2 I)^{-1}\mathbf{k}(\mathbf{d}),$$

where $\mathbf{D}_t$ and $\mathbf{y}$ are the descriptors and observed performances of the gaits tried so far. The next gait maximises the upper confidence bound $\mu_t + \kappa\sigma_t$; ITE stops when the best observed performance reaches $\alpha \max_{\mathbf{d}} \mu_t(\mathbf{d})$. One trial is *one* episode on the damaged robot.

**Prior** The fitness stored in a repertoire comes from a single noisy episode, and MAP-Elites keeps the luckiest evaluations: elites are over-estimated (on development repertoires, the correlation between stored and re-evaluated fitness is 0.88, and the "best" cell of a MAP-Elites repertoire performs worse than a standing robot once re-evaluated). As the repertoire is built offline in simulation, we re-evaluate it on the intact robot (4 episodes) and use this as prior; the best-intact baseline uses the same values.

**Hyper-parameters** With the values of Cully et al. [5] (length-scale 0.4, unit signal variance, noise $10^{-3}$), ITE stopped after very few trials and plateaued. The residuals have a variance of 0.03–0.06, not 1, and a large nugget (neighbouring cells differ). We therefore fitted $(\ell, \sigma_f^2, \sigma_n^2)$ by maximising the marginal likelihood on *development* repertoires (short runs of the three algorithms, never used for the results): $\ell = 0.4$, $\sigma_f^2 = 0.05$, $\sigma_n^2 = 0.02$ (log-likelihood $+4059$ against $-1280$ for the original values). $\kappa = 0.05$ and $\alpha = 0.9$ are those of Cully et al. [5]; values of $\kappa$ from 0.05 to 2 made little difference on the development set.

### 4.3 Baselines

*Best-intact*: replay the best intact gait (no adaptation, one trial). *Top-k*: try cells by decreasing prior, without a model. *Random*: try cells at random. All methods run for 20 trials and are repeated 5 times per (repertoire, damage); the quality of the recommended gait (best observed so far) is measured with its oracle mean.

### 4.4 Statistics

The repetitions of an adaptation only differ by episode noise, so they are averaged first: the statistical unit is a (repertoire, damage) pair, i.e. 36 units per algorithm. Algorithms are compared with two-sided Mann-Whitney tests on these units, and two adaptation methods with paired two-sided Wilcoxon signed-rank tests. Units of the same seed share a repertoire and are not independent. As a check that accounts for this, we also fit linear mixed models with a random intercept per repertoire, `value ~ algorithm + damage + (1 | repertoire)` for the comparisons between algorithms and `difference ~ algorithm + (1 | repertoire)` for the paired comparisons between methods, and report differences with 95% confidence intervals (Section 5.7). With 3 repertoires per algorithm these intervals are approximate. Medians in the tables are taken over all repetitions. Throughout, $v$ denotes the normalised performance and $p$ a p-value.

### 4.5 RL re-training

TD3 (QDax implementation) is trained on the damaged robot, either from scratch or starting from the best intact DCRL-ME policy (same architecture; the critic starts from scratch), on 4 damages (FR leg paralysed, HL leg weakened, FL+FR legs paralysed, FL ankle paralysed) and 2 seeds, for 2000 episodes with 16 robots in parallel and one gradient step per transition. The deterministic policy is evaluated on 8 episodes after every 16 training episodes. Fine-tuning starts from the best re-evaluated elite. Because the new critic is untrained at first, we also test two remedies: a *warm-up*, where the actor is frozen during the first 160 episodes on the damaged robot (counted), and a *pre-trained critic*, trained for 512 episodes on the intact robot in simulation with the actor frozen (not counted, since no damaged robot is involved).

**Damage-robust policy** The strongest alternative to adaptation is not to adapt at all: train one policy that works under many damages. We train TD3 from scratch in simulation for 10,000 episodes, drawing a new damage at every episode (each joint independently paralysed or weakened with probability 0.25), and evaluate it without any trial on the damaged robot (32

episodes per damage). The test damages lie inside this training distribution, which favours the robust policy: unlike ITE, it needs to know in advance what kind of damage can happen.

# 5 Results

## 5.1 Repertoires on the intact robot

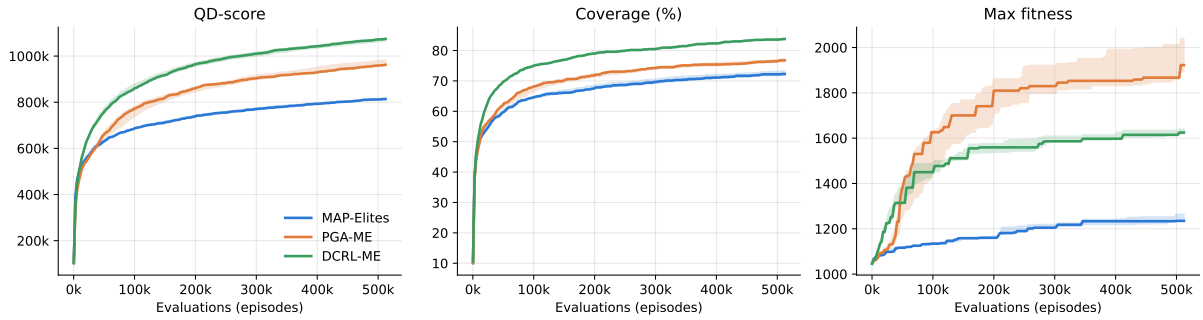


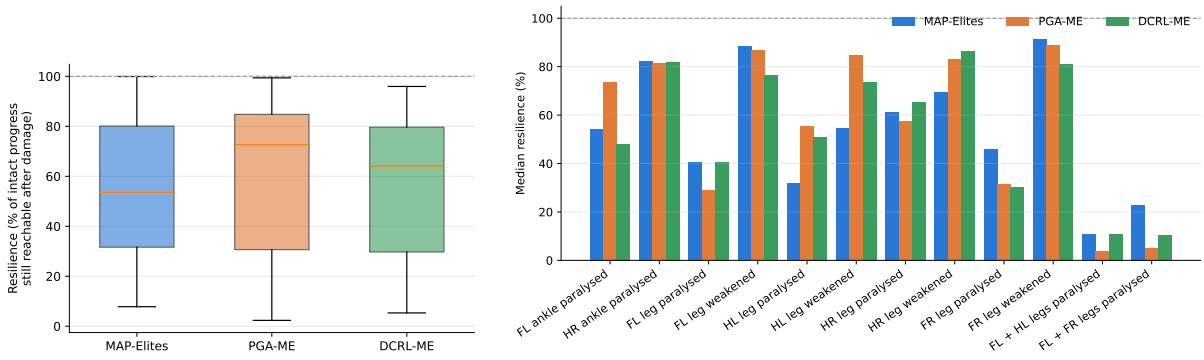

**Figure 2:** QD-score, coverage and best fitness during training (median and inter-quartile range over 3 seeds).

**Table 2:** Repertoires at the end of training (median over seeds).

| Algorithm | Coverage (%) | Max fitness | QD-score ($\times 10^3$) |
|---|---|---|---|
| MAP-Elites | 72.3 | 1235 | 814 |
| PGA-ME | 76.8 | 1922 | 962 |
| DCRL-ME | 83.8 | 1625 | 1074 |

Both RL-based algorithms improve over MAP-Elites at equal budget (Figure 2 and Table 2), in different ways. DCRL-ME fills the most cells and reaches the highest QD-score, which is consistent with its descriptor-conditioned critic improving gaits *without changing their style*. PGA-ME finds by far the fastest gaits (median best fitness 1922 against 1625 for DCRL-ME and 1235 for MAP-Elites): its unconditioned critic pushes offspring towards high reward, wherever they land in the grid. The fastest gait we inspected (PGA-ME) is a bounding gait in which the feet touch the ground only briefly. The stored fitness of these elites is optimistic (Section 6).

## 5.2 Resilience to damage (Q1)




**(a)** All damages

**(b)** By damage (median over seeds)

**Figure 3:** Resilience: share of the intact oracle progress still reachable after damage.

**Table 3:** Resilience (%) and absolute oracle performance $v$ after damage (median over seeds).

| Damage | MAP-Elites (%) | PGA-ME (%) | DCRL-ME (%) | MAP-Elites ($v$) | PGA-ME ($v$) | DCRL-ME ($v$) |
|---|---|---|---|---|---|---|
| FL ankle paralysed | 54 | 74 | 48 | 0.28 | 1.86 | 0.83 |
| HR ankle paralysed | 82 | 81 | 82 | 0.35 | 2.06 | 1.48 |
| FL leg paralysed | 41 | 29 | 41 | 0.21 | 0.73 | 0.73 |
| FL leg weakened | 88 | 87 | 76 | 0.38 | 2.20 | 1.32 |
| HL leg paralysed | 32 | 55 | 51 | 0.13 | 1.72 | 0.92 |
| HL leg weakened | 55 | 85 | 74 | 0.24 | 2.33 | 1.32 |
| HR leg paralysed | 61 | 58 | 65 | 0.23 | 1.79 | 1.18 |
| HR leg weakened | 69 | 83 | 87 | 0.28 | 2.15 | 1.56 |
| FR leg paralysed | 46 | 31 | 30 | 0.20 | 0.97 | 0.54 |
| FR leg weakened | 91 | 89 | 81 | 0.40 | 2.25 | 1.42 |
| FL + HL legs paralysed | 11 | 4 | 11 | 0.06 | 0.10 | 0.19 |
| FL + FR legs paralysed | 23 | 5 | 10 | 0.12 | 0.16 | 0.18 |

In relative terms, the three kinds of repertoires are equally resilient: the median share of the intact progress that remains reachable after damage is 54% for MAP-Elites, 73% for PGA-ME and 64% for DCRL-ME, and none of the pairwise differences is significant (Mann-Whitney, $p = 0.532$, $p = 0.996$ and $p = 0.496$; Figure 3). We therefore find no sign that RL makes repertoires less useful after damage. In absolute terms, the difference is large (Table 3): after damage, the best gait available is worth a median $v = 1.84$ in PGA-ME repertoires, $v = 1.14$ in DCRL-ME ones and only $v = 0.23$ in MAP-Elites ones. Weakening a leg is usually mild; paralysing a whole leg removes roughly half of the progress; with two paralysed legs, no repertoire keeps a useful gait.

### 5.3 Adaptation in a few trials (Q2)

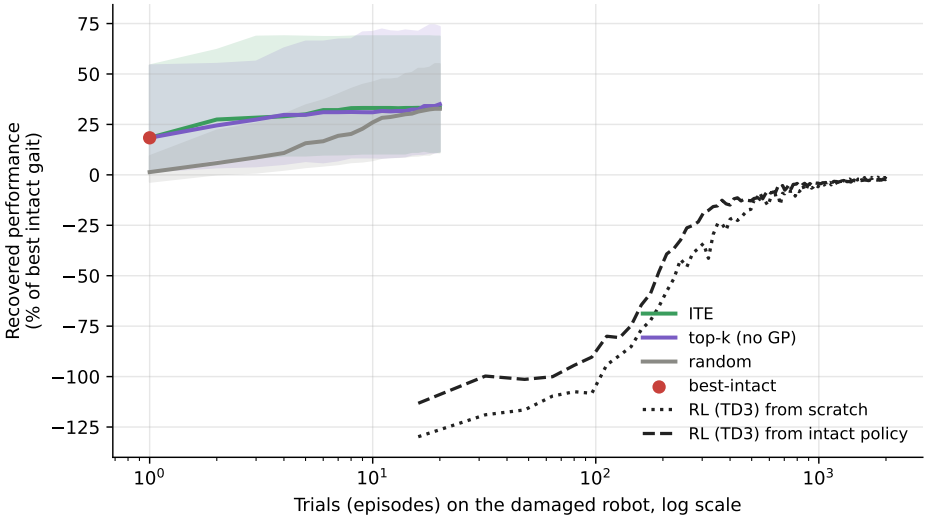



**Figure 4:** Performance of the recommended gait against the number of trials on the damaged robot (median and IQR over repertoires, damages and repetitions; each repertoire relative to its own best intact gait). RL curves are relative to the best intact DCRL-ME gait.

ITE meets its stopping criterion in 89% of the runs, after a median of 5 trials (IQR 2–10). The recommended gait recovers 33% of the progress of the best intact gait, against 18% when the best intact gait is simply replayed, and 62% of what an exhaustive search of the repertoire would find (Table 4). Random search is much slower (9% after three trials, against 28% for ITE).

The comparison with top-$k$, which tries gaits in decreasing order of intact performance without any model, is less favourable: after three trials ITE reaches 28% and top-$k$ 27%; after 20 trials, 33% and 35% (Figure 4 and Table 5). Paired over the 108 (repertoire, damage) units, ITE is better than top-$k$ after five trials in 47% of them, equal in 5% and worse in 48% (Wilcoxon

**Table 4:** Adaptation methods (median over repertoires, damages and repetitions). "% intact": share of the progress of the best intact gait; "% oracle": share of the best gait available after damage.

| Method | Trials | % intact | % oracle | Absolute $v$ |
|---|---|---|---|---|
| ITE | 5 | 33 | 62 | 0.37 |
| top-$k$ (no GP) | 20 | 35 | 77 | 0.40 |
| random | 20 | 33 | 60 | 0.31 |
| best-intact | 1 | 18 | 34 | 0.19 |

**Table 5:** Share of the intact progress (%) recovered after a given number of trials (median).

| Method | 1 trial | 3 trials | 5 trials | 10 trials | 20 trials |
|---|---|---|---|---|---|
| ITE | 18 | 28 | 30 | 33 | 33 |
| top-$k$ (no GP) | 18 | 27 | 30 | 31 | 35 |
| random | 1 | 9 | 16 | 26 | 33 |

$p = 0.445$). After three trials ITE has a small advantage (mean difference +0.05 in $v$, $p = 0.009$), which comes mostly from DCRL-ME repertoires (+0.13, $p = 0.019$) and disappears after five and ten trials (Table 6). The pooled medians of Table 7 suggest a larger gap on DCRL-ME repertoires (0.59 against 0.48 after ten trials), but the paired test does not support it ($p = 0.407$): unit by unit, ITE is better as often as it is worse (Table 6). On PGA-ME repertoires, the best intact gait already keeps most of its performance ($v = 1.43$ at the first trial). On MAP-Elites repertoires, all methods stay close to a standing robot.

**Table 6:** ITE against top-$k$ at equal numbers of trials: mean difference in absolute performance $v$ (ITE − top-$k$), number of units where ITE is better / equal / worse, and paired Wilcoxon test, over the (repertoire, damage) units.

| Repertoire | 3 trials | 5 trials | 10 trials |
|---|---|---|---|
| MAP-Elites | +0.01 (18/0/18; $p = 0.581$) | +0.00 (16/0/20; $p = 0.798$) | +0.01 (20/0/16; $p = 0.451$) |
| PGA-ME | +0.03 (19/4/13; $p = 0.135$) | −0.03 (18/1/17; $p = 0.974$) | −0.03 (18/0/18; $p = 0.944$) |
| DCRL-ME | +0.13 (20/1/15; $p = 0.019$) | +0.08 (17/4/15; $p = 0.166$) | +0.04 (17/2/17; $p = 0.407$) |

**Table 7:** Absolute performance $v$ of the recommended gait at equal numbers of trials, by repertoire (median).

| Repertoire – method | 1 trial | 3 trials | 5 trials | 10 trials | 20 trials |
|---|---|---|---|---|---|
| MAP-Elites – ITE | 0.07 | 0.10 | 0.11 | 0.11 | 0.12 |
| MAP-Elites – top-$k$ | 0.07 | 0.09 | 0.10 | 0.09 | 0.11 |
| MAP-Elites – random | -0.03 | 0.02 | 0.07 | 0.09 | 0.12 |
| PGA-ME – ITE | 1.43 | 1.47 | 1.47 | 1.47 | 1.47 |
| PGA-ME – top-$k$ | 1.43 | 1.47 | 1.52 | 1.52 | 1.57 |
| PGA-ME – random | 0.10 | 0.21 | 0.47 | 0.78 | 1.11 |
| DCRL-ME – ITE | 0.22 | 0.47 | 0.47 | 0.59 | 0.60 |
| DCRL-ME – top-$k$ | 0.22 | 0.36 | 0.37 | 0.48 | 0.59 |
| DCRL-ME – random | 0.03 | 0.22 | 0.28 | 0.49 | 0.59 |

What a robot finally gets depends mostly on how its repertoire was built (Table 8): with ITE, the recommended gait is worth $v = 1.47$ with PGA-ME, $v = 0.59$ with DCRL-ME and $v = 0.11$ with MAP-Elites (Mann-Whitney on the (repertoire, damage) units: $p < 0.001$ for MAP-Elites against PGA-ME, $p < 0.001$ against DCRL-ME and $p = 0.002$ between PGA-ME and DCRL-ME; at the level of seeds, every PGA-ME repertoire is above every DCRL-ME repertoire, itself above

**Table 8:** ITE on each kind of repertoire (median over seeds, damages and repetitions).

| Repertoire | Trials | % intact | Absolute $v$ |
|---|---|---|---|
| MAP-Elites | 6 | 24 | 0.11 |
| PGA-ME | 3 | 55 | 1.47 |
| DCRL-ME | 6 | 33 | 0.59 |

every MAP-Elites repertoire). Figures 5 and 6 show one ITE run.

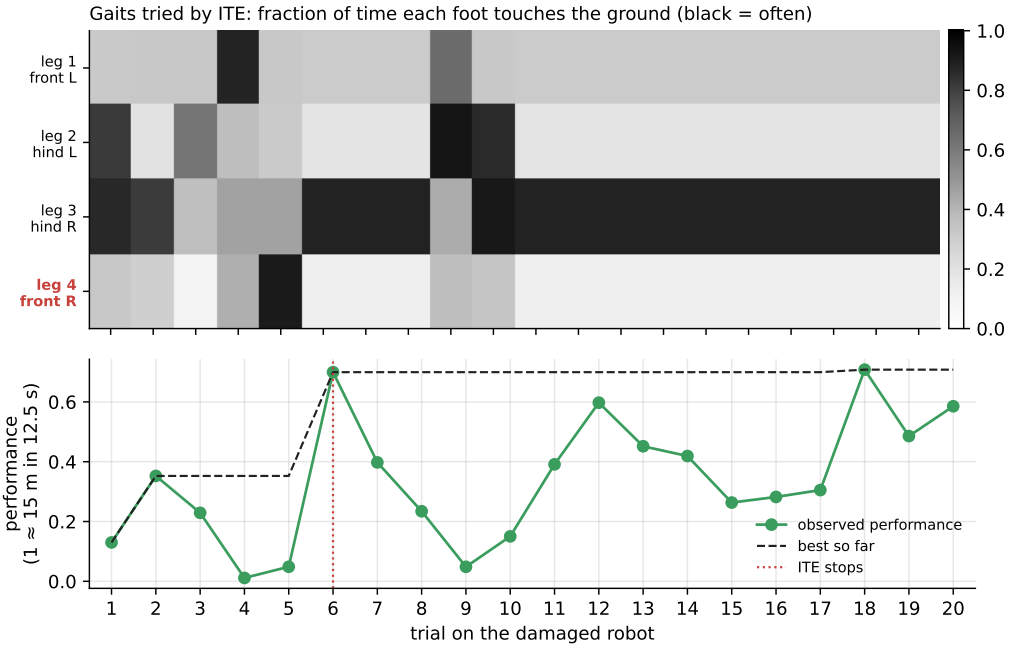



**Figure 5:** One ITE run on a DCRL-ME repertoire (front-right leg paralysed; the five repetitions all recommend the same gait, after 6 or 7 trials, and this one used 6): feet-contact profile of each gait tried, and its observed performance.

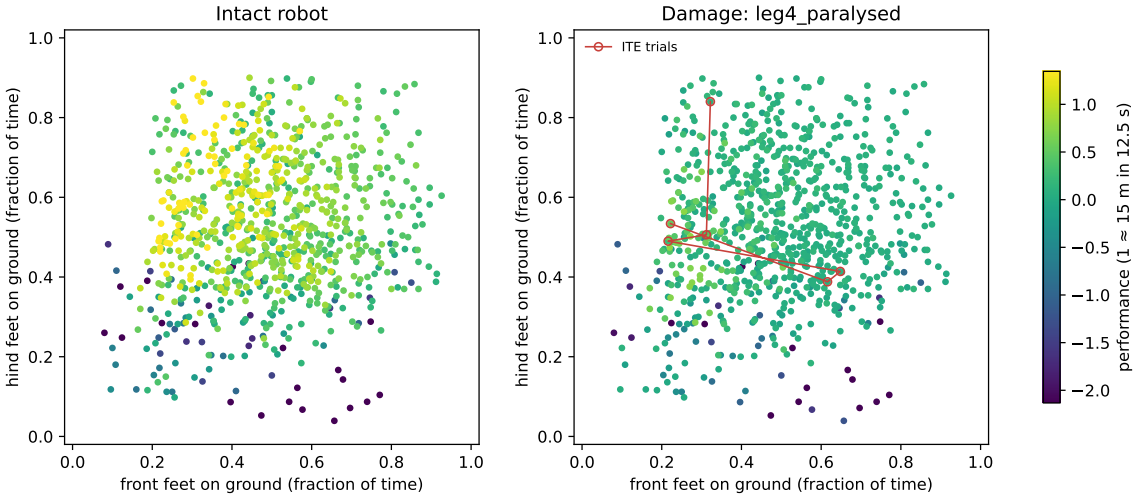



**Figure 6:** The same repertoire before and after damage (front vs hind feet contact), and the gaits tried by ITE.

## 5.4 Comparison with RL re-training (Q3)

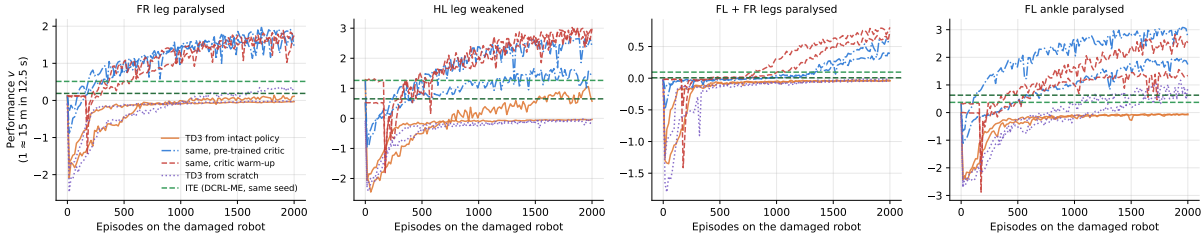



**Figure 7:** TD3 on the damaged robot (two seeds each): from scratch, from the best intact DCRL-ME policy, and from the same policy with a critic warm-up or a pre-trained critic, against the performance of the gait found by ITE on the DCRL-ME repertoire of the same seed (dashed lines).

**Table 9:** Episodes on the damaged robot needed by each TD3 variant to match the gait found by ITE (DCRL-ME repertoire of the same seed, mean number of ITE trials in the first column); "–": not reached within 2000 episodes.

| Damage | ITE trials | Scratch | Fine-tune | Warm-up | Pre-trained critic |
|---|---|---|---|---|---|
| FL ankle paralysed, seed 0 | 10.0 | 1072 | – | 496 | 112 |
| FL ankle paralysed, seed 1 | 5.6 | 1808 | – | 400 | 576 |
| HL leg weakened, seed 0 | 1.4 | – | 0 | 0 | 1088 |
| HL leg weakened, seed 1 | 6.4 | – | 1536 | 336 | 224 |
| FR leg paralysed, seed 0 | 6.6 | – | – | 496 | 192 |
| FR leg paralysed, seed 1 | 15.0 | 1264 | – | 224 | 192 |
| FL + FR legs paralysed, seed 0 | 18.8 | – | – | 1008 | 1344 |
| FL + FR legs paralysed, seed 1 | 20.0 | – | – | 720 | 720 |

Within 2000 episodes, TD3 from scratch matches the gait found by ITE in 3/8 runs (median 1264 episodes when it does), and plain fine-tuning in only 2/8 runs, one of them because the intact policy already matched ITE before any training (Figure 7 and Table 9). Plain fine-tuning collapses at the first evaluation. The two remedies show why: with a critic warm-up, fine-tuning matches ITE in 8/8 runs (median 448 episodes), and with a critic pre-trained on the intact robot in 8/8 runs (median 400 episodes). The collapse came from following the gradient of an

untrained critic. Even so, the best re-training still needs hundreds of episodes on the damaged robot, against 1 to 20 trials for ITE.

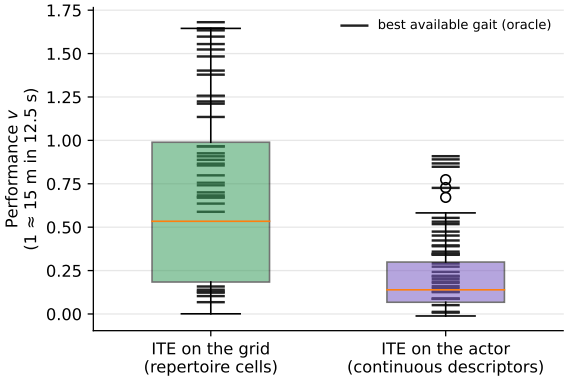
### 5.5 A damage-robust policy

**Table 10:** Performance $v$ after damage of the damage-robust TD3 policy (zero trials, mean of 2 seeds) and of the gait found by ITE (mean over seeds and repetitions), on the nominal robot and with the reality gap of Section 5.7.

| Damage | Robust TD3 | ITE, DCRL-ME | ITE, PGA-ME | Robust TD3 (gap) | ITE, PGA-ME (gap) |
|---|---|---|---|---|---|
| FL ankle paralysed | 1.29 | 0.50 | 1.35 | 1.38 | 1.05 |
| HR ankle paralysed | 2.21 | 1.09 | 2.05 | 1.99 | 1.64 |
| FL leg paralysed | 0.55 | 0.39 | 0.48 | 0.73 | 0.32 |
| FL leg weakened | 2.09 | 0.87 | 1.96 | 1.91 | 1.67 |
| HL leg paralysed | 0.75 | 0.35 | 1.66 | 0.48 | 1.44 |
| HL leg weakened | 1.89 | 0.86 | 2.26 | 1.75 | 2.00 |
| HR leg paralysed | 1.63 | 0.59 | 0.91 | 1.43 | 0.87 |
| HR leg weakened | 2.25 | 1.13 | 1.90 | 2.08 | 1.64 |
| FR leg paralysed | 0.63 | 0.36 | 0.71 | 0.48 | 0.63 |
| FR leg weakened | 2.26 | 1.20 | 2.25 | 1.99 | 1.92 |
| FL + HL legs paralysed | -0.14 | 0.06 | 0.03 | -0.21 | 0.01 |
| FL + FR legs paralysed | -0.48 | 0.03 | 0.04 | -0.36 | 0.06 |

The robust policy changes the picture (Table 10). Without any trial on the damaged robot it reaches a median $v = 1.46$ over the damages, better than ITE on DCRL-ME repertoires in 10 of the 12 damages and as good as ITE on the best repertoires, those of PGA-ME (better in 6 of 12; median $v = 1.51$ for ITE). It is also a strong intact gait ($v = 2.06$) and is little affected by the reality gap ($v = 1.40$ against 1.24 for ITE on PGA-ME). It fails, however, when two legs are paralysed, where it does worse than a standing robot, while the repertoires still offer a gait that barely moves. Two caveats limit this comparison: the test damages belong to the training distribution of the robust policy, and ITE assumes nothing about the damage. A damage outside that distribution, or a real robot, could reverse the ranking; this is precisely the case ITE was designed for.

### 5.6 Bonus: searching the continuous descriptor space




| ITE search space | Trials | Found $v$ | Oracle $v$ |
|---|---|---|---|
| Grid (repertoire cells) | 6 | 0.53 | 0.90 |
| Continuous (conditioned actor) | 3 | 0.14 | 0.32 |

**Figure 8:** ITE on the cells of a DCRL-ME repertoire against ITE on 2048 gaits generated by its descriptor-conditioned actor for targets spread over $[0, 1]^4$ (3 dedicated runs, same runs for both).

DCRL-ME also learns an actor $\pi(a \mid o, \mathbf{d})$ that should produce a gait with any requested feet-contact profile $\mathbf{d}$. We let ITE search 2048 such gaits instead of the repertoire cells, on 3 dedicated DCRL-ME runs where the actor was saved. The answer is clearly negative (Figure 8): the gaits produced by the actor are much weaker than the elites (best available gait after damage

$v = 0.32$ against $v = 0.90$, and never better in any of the (run, damage) pairs), and ITE on the actor finds $v = 0.14$ against $v = 0.53$ on the grid (better in only 3 of the 36 (run, damage) pairs; Wilcoxon $p < 0.001$). The actor is a useful source of offspring during training, but it does not replace the elites that were selected and refined.

### 5.7 Robustness of the conclusions

**Oracle bias** Re-evaluating the cell chosen by the oracle on 16 fresh episodes shows that the oracle over-estimates the best gait by $v \approx 0.06$ (MAP-Elites), 0.08 (PGA-ME) and 0.06 (DCRL-ME) on average (Table 11). With these unbiased values the median resilience is 50%, 66% and 60%; the mixed model now finds PGA-ME slightly more resilient than MAP-Elites (difference $+0.08$ $[+0.01, +0.16]$ with the correction, $p = 0.030$, against $+0.04$ $[-0.04, +0.12]$ without it), but the rank test does not. Either way, building the repertoire with RL does not make it less resilient.

**Table 11:** Oracle re-evaluated on fresh episodes: mean bias of the 4-episode oracle, and medians of the resilience and of the best gait after damage computed with the re-evaluated values.

| Repertoire | Bias of the oracle ($v$) | Resilience, checked (%) | Best gait after damage, checked ($v$) |
|---|---|---|---|
| MAP-Elites | 0.06 | 50 | 0.17 |
| PGA-ME | 0.08 | 66 | 1.68 |
| DCRL-ME | 0.06 | 60 | 1.04 |

**Mixed models** Linear mixed models with a random intercept per repertoire confirm the comparisons between algorithms (Table 12): no significant difference in relative resilience, large differences in absolute performance. For ITE against top-$k$ (Table 13), they agree with the Wilcoxon tests on MAP-Elites and PGA-ME, but on DCRL-ME they find a positive mean gain after three and five trials ($+0.08$ $[+0.02, +0.15]$ after five trials, $p = 0.014$) while the median gain is zero: the model helps a lot on a few damages and not at all on the others.

**Table 12:** Differences between algorithms from linear mixed models (fixed effects: algorithm and damage; random intercept per repertoire), with 95% confidence intervals.

| Quantity | Comparison | Difference [95% CI] | $p$-value |
|---|---|---|---|
| Resilience | PGA-ME − MAP-Elites | $+0.04$ $[-0.04, +0.12]$ | 0.357 |
| Resilience | DCRL-ME − MAP-Elites | $+0.01$ $[-0.08, +0.09]$ | 0.888 |
| Resilience | DCRL-ME − PGA-ME | $-0.03$ $[-0.11, +0.05]$ | 0.435 |
| Best gait after damage ($v$) | PGA-ME − MAP-Elites | $+1.31$ $[+1.10, +1.51]$ | <0.001 |
| Best gait after damage ($v$) | DCRL-ME − MAP-Elites | $+0.74$ $[+0.53, +0.94]$ | <0.001 |
| Best gait after damage ($v$) | DCRL-ME − PGA-ME | $-0.57$ $[-0.78, -0.37]$ | <0.001 |
| Gait found by ITE ($v$) | PGA-ME − MAP-Elites | $+1.16$ $[+0.97, +1.36]$ | <0.001 |
| Gait found by ITE ($v$) | DCRL-ME − MAP-Elites | $+0.48$ $[+0.29, +0.68]$ | <0.001 |
| Gait found by ITE ($v$) | DCRL-ME − PGA-ME | $-0.68$ $[-0.88, -0.48]$ | <0.001 |

**Table 13:** Mean gain of ITE over top-$k$ in $v$ (95% confidence interval) from a mixed model per repertoire type.

| Repertoire | 3 trials | 5 trials | 10 trials |
|---|---|---|---|
| MAP-Elites | $+0.01$ $[-0.04, +0.06]$ | $+0.00$ $[-0.06, +0.07]$ | $+0.01$ $[-0.07, +0.09]$ |
| PGA-ME | $+0.03$ $[-0.02, +0.08]$ | $-0.03$ $[-0.09, +0.04]$ | $-0.03$ $[-0.11, +0.05]$ |
| DCRL-ME | $+0.13$ $[+0.08, +0.18]$ | $+0.08$ $[+0.02, +0.15]$ | $+0.04$ $[-0.04, +0.12]$ |

**Is the descriptor the bottleneck?** With the feet-contact descriptor the fitted noise variance (0.020) is not small compared with the signal variance (0.050): neighbouring cells react differently to the same damage. A damage acts on joints, so we built a second descriptor, the mean absolute action of each of the 8 actuators on the intact robot, and fitted the Gaussian process on it in the same way, on the same development repertoires (Table 14). It explains the residuals much better: the log-likelihood gained over independent cells is 152.2 per (repertoire, damage) pair against 51.7, and the fitted noise drops to 0.003. Yet ITE barely improves (Table 15): against ITE on feet contact the mean gain is +0.02 after five trials ($p = 0.923$) and +0.03 after ten ($p = 0.757$). Against top-$k$, ITE on joint usage is ahead after three trials (+0.07, $p = 0.013$) and, on DCRL-ME, after five (+0.13, $p = 0.015$), but not after ten. These runs were made on a CPU, so ITE and top-$k$ were re-run on the same machine for these comparisons. A better model of the damage therefore does not translate into a better choice of gait: in these repertoires, the gaits that remain good after a damage are mostly among the best intact ones, which top-$k$ tries first anyway.

**Table 14:** Gaussian-process hyper-parameters fitted by maximum marginal likelihood on the development repertoires for each descriptor, and log-likelihood gained over independent cells, per (repertoire, damage) pair.

| Descriptor | Length-scale | Signal var. | Noise var. | Gain per pair |
|---|---|---|---|---|
| feet contact (4D) | 0.4 | 0.050 | 0.020 | 51.7 |
| joint usage (8D) | 0.2 | 0.100 | 0.003 | 152.2 |

**Table 15:** ITE on joint usage minus ITE on feet contact: mean difference in $v$ and paired Wilcoxon test, over the (repertoire, damage) units.

| Repertoire | 3 trials | 5 trials | 10 trials |
|---|---|---|---|
| MAP-Elites | −0.01 ($p = 0.023$) | −0.01 ($p = 0.369$) | −0.01 ($p = 0.062$) |
| PGA-ME | +0.06 ($p = 0.445$) | +0.03 ($p = 0.808$) | +0.03 ($p = 0.793$) |
| DCRL-ME | +0.01 ($p = 0.728$) | +0.04 ($p = 0.351$) | +0.06 ($p = 0.265$) |

**Reality gap** Here the simulator also plays the robot: prior and trials come from the same physics, which is the easy case. To mimic a deployment, the trials now run on a robot whose physics differ from the simulator that built the repertoire and gave the prior: ground friction ×0.7, torso mass ×1.2, actuator strength ×0.9, values fixed before the experiment. The gap alone costs little: the best intact gait keeps 91% of its progress. With a damage on top of it (Table 16), the gait found by ITE is worth $v = 0.28$ after a median of 6 trials, top-$k$ reaches 0.27 after 20 trials, and the best intact gait 0.10. ITE's early lead over top-$k$ is slightly larger than without the gap: +0.06 after three trials ($p = 0.002$), +0.16 on DCRL-ME ($p = 0.003$); but it is gone after five trials (+0.03, $p = 0.390$), and on PGA-ME top-$k$ is ahead after ten (−0.08, $p = 0.009$). A moderate gap therefore makes the prior less reliable without changing the conclusion: the model mostly helps in the first trials.

**Table 16:** Adaptation with a reality gap: prior from the nominal simulator, trials on the perturbed robot (medians over repertoires, damages and repetitions; "% intact": share of the progress of the best intact gait of the perturbed robot).

| Method | Trials | Nominal $v$ | With reality gap $v$ | % intact (gap) |
|---|---|---|---|---|
| best-intact | 1 | 0.19 | 0.10 | 12 |
| top-$k$ | 20 | 0.40 | 0.27 | 33 |
| ITE | 6 | 0.37 | 0.28 | 26 |

# 6 Discussion

**Why ITE is not much better than top-$k$** ITE helps when trying one gait tells something about *similar* gaits, and when the best gait after damage is not among the first ones a ranking would try. With the feet-contact descriptor the first condition is weak: neighbouring cells react differently to the same damage. The descriptor ablation (Section 5.7) shows that a descriptor based on joint usage fixes this, with a noise variance seven times smaller, but ITE still gains little: the second condition fails. A damaged repertoire mostly keeps some of its best gaits, so the re-evaluated intact ranking is already a strong guide, and 20 trials are enough for top-$k$ to reach them. ITE should matter more when the prior is misleading, for instance when the simulator and the robot differ (Section 5.7), or for damages that remove the best gaits altogether.

**Noisy elites** A repertoire stores the fitness of a single noisy episode, and MAP-Elites keeps the luckiest ones: elites are over-estimated. Re-evaluating the repertoire before damage, offline in simulation, is cheap and was necessary for both ITE and the best-intact baseline. This is the problem studied by uncertain QD.

**The oracle is optimistic** The oracle is the maximum of noisy means over $\sim$1000 cells, so it is biased upwards; percentages of the oracle under-estimate all methods.

**Counting episodes, not compute** In simulation, RL consumes thousands of episodes in minutes; on a real robot, each episode costs time, wear and supervision. We therefore compare trials and episodes, not wall-clock time.

# 7 Limitations

Simulation only, one robot, and damages modelled as action scaling; the reality gap is simulated by changing a few physical parameters. One repertoire size and one budget, with the hyper-parameters of the official notebooks. 3 seeds per algorithm, so the (repertoire, damage) units used in the tests are not independent and the p-values are indicative, even with mixed models. ITE hyper-parameters fitted on development repertoires of the same algorithms. RL baselines on 4 damages and 2 seeds; the robust policy is trained on the family of damages it is tested on, and no meta-learned or history-based adaptive policy [11, 13] was tested. Results on GPU and CPU differ slightly for the same seed (different floating-point rounding), so ITE with the joint-usage descriptor is compared with ITE and top-$k$ re-run on the same CPU.

# 8 Conclusion and future work

Repertoires built with the help of RL critics are not less useful after damage: they are as resilient in relative terms and keep much better gaits in absolute terms, so that the quality of the repertoire, more than the adaptation algorithm, decides how well a damaged robot walks. ITE finds a good gait in a handful of trials, far fewer than RL re-training, but in this setting most of its benefit comes from a good prior: a simple ranking of re-evaluated gaits is almost as good, and a better model of the damage does not change this. A single policy trained on random damages in simulation is a serious competitor when the possible damages are known in advance. The case for repertoires and ITE is therefore the unexpected damage on a real robot, which is where future work should test them, together with damages outside the training distribution of robust policies and meta-learned adaptive policies.

# References


[1] Maxime Allard, Simón C. Smith, Konstantinos Chatzilygeroudis, Bryan Lim, and Antoine Cully. Online damage recovery for physical robots with hierarchical quality-diversity. *ACM Transactions on Evolutionary Learning and Optimization*, 2023.

[2] Félix Chalumeau, Raphael Boige, Bryan Lim, Valentin Macé, Maxime Allard, Arthur Flajolet, Antoine Cully, and Thomas Pierrot. Neuroevolution is a competitive alternative to reinforcement learning for skill discovery. In *International Conference on Learning Representations (ICLR)*, 2023.

[3] Félix Chalumeau, Bryan Lim, Raphael Boige, Maxime Allard, Luca Grillotti, Manon Flageat, Valentin Macé, Arthur Flajolet, Thomas Pierrot, and Antoine Cully. QDax: A library for quality-diversity and population-based algorithms with hardware acceleration. *arXiv preprint arXiv:2308.03665*, 2023.

[4] Konstantinos Chatzilygeroudis, Vassilis Vassiliades, and Jean-Baptiste Mouret. Reset-free trial-and-error learning for robot damage recovery. *Robotics and Autonomous Systems*, 100:236–250, 2018.

[5] Antoine Cully, Jeff Clune, Danesh Tarapore, and Jean-Baptiste Mouret. Robots that can adapt like animals. *Nature*, 521(7553):503–507, 2015.

[6] Maxence Faldor, Félix Chalumeau, Manon Flageat, and Antoine Cully. Synergizing quality-diversity with descriptor-conditioned reinforcement learning. *ACM Transactions on Evolutionary Learning and Optimization*, 2024. arXiv:2401.08632.

[7] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning (ICML)*, 2017.

[8] Manon Flageat and Antoine Cully. Uncertain quality-diversity: Evaluation methodology and new methods for quality-diversity in uncertain domains. *IEEE Transactions on Evolutionary Computation*, 2023.

[9] Scott Fujimoto, Herke Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *International Conference on Machine Learning (ICML)*, pages 1587–1596, 2018.

[10] Rituraj Kaushik, Pierre Desreumaux, and Jean-Baptiste Mouret. Adaptive prior selection for repertoire-based online adaptation in robotics. *Frontiers in Robotics and AI*, 6, 2020.

[11] Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. In *Robotics: Science and Systems (RSS)*, 2021.

[12] Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.

[13] Anusha Nagabandi, Ignasi Clavera, Simin Liu, Ronald S. Fearing, Pieter Abbeel, Sergey Levine, and Chelsea Finn. Learning to adapt in dynamic, real-world environments through meta-reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2019.

[14] Olle Nilsson and Antoine Cully. Policy gradient assisted MAP-Elites. In *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO)*, pages 866–875, 2021.

[15] Xue Bin Peng, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. Sim-to-real transfer of robotic control with dynamics randomization. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2018.

[16] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017.

[17] Wenhao Yu, C. Karen Liu, and Greg Turk. Policy transfer with strategy optimization. In *International Conference on Learning Representations (ICLR)*, 2019.

# A Reproducibility

```
python -m qd_damage.repertoires -algo {me,pgame,dcrlme} -seed S
python -m qd_damage.oracle -runs results/repertoires/<run>
python -m qd_damage.oracle -check-best -runs results/repertoires/<run>
python -m qd_damage.adaptation -runs results/repertoires/<run>
python -m qd_damage.calibrate [-joint-usage] -oracles results/oracle/*_demo
python -m qd_damage.joint_usage -runs results/repertoires/<run>
python -m qd_damage.adaptation -joint-usage -methods ite ...
python -m qd_damage.oracle -reality-gap -out results/oracle_gap -runs ...
python -m qd_damage.adaptation -reality-gap -oracle-root results/oracle_gap ...
python -m qd_damage.rl_retrain -scenario <s> -variant <variant> -seed S
python -m qd_damage.rl_retrain -robust -seed S
python -m qd_damage.analysis.report
```

Versions: QDax 0.5.0, JAX 0.4.28, Brax 0.10.4 (pinned; see `requirements-lock.txt`). The GPU runs were made on Kaggle (2× Tesla T4) with the scripts of `kaggle/`; the oracle check, the joint-usage descriptor and its CPU control on a laptop CPU. Every run stores its configuration, seed, code commit, library versions and device.